## INTRO TO STATISTICS

It's all about interpreting data (Table, list, graphs, etc)

Data can be form

**Population** (the entire list of a specified group)

**Mean** (a subset of the Population)

We usually investigate a small sample of the population to draw conclusions for the whole population itself.

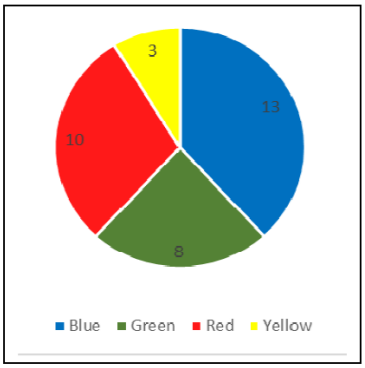
| Discrete | OR | Continuous |
|---|---|---|
| $\{10, 20, 30\}$ | | $[40, 100]$ |
| $\{0, 1, 2, 3, \ldots\}$ | | $\mathbb{R}$ |
| (finite or numerable set) | | (interval) |

Data can be organized in several ways. We present some examples below

Frequency table

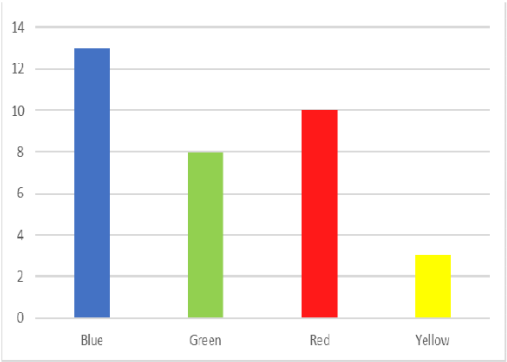
| Colored Balls | Frequency |
|---|---|
| Blue | 13 |
| Green | 8 |
| Red | 10 |
| Yellow | 3 |

Pie chart



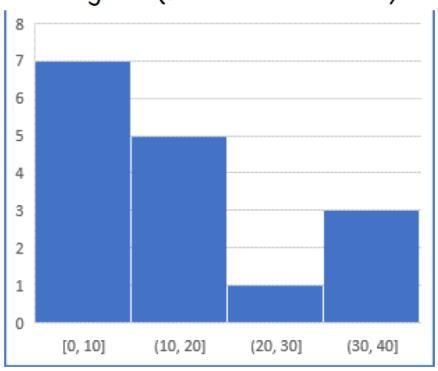
| Colored Balls | Freq |
|---|---|
| Blue | 13 |
| Green | 8 |
| Red | 10 |
| Yellow | 3 |

Bar graph (for discrete data)




| Age | Frequency |
|---|---|
| [0,10) | 7 |
| [10,20) | 5 |
| [20,30) | 1 |
| [30,40) | 3 |

Histogram (for continuous data)




Stem and leaf Diagram

| Data |
|---|
| 12, 14, 16, 16, 20, 21 |
| 21, 21, 25, 32, 39, 40 |
| 43, 44, 47, 48, 49, 53 |

Key: 1|3 represents 13

| Stem | Leaf |
|---|---|
| 1 | 2, 4, 6, 6 |
| 2 | 0, 1, 1, 1, 5 |
| 3 | 2, 9 |
| 4 | 0, 3, 4, 7, 8, 9 |
| 5 | 3 |

Ex: Age of Students in class:

16,15,17,15,15,16,18,15

Mean:

Notation:

Example: The data below is listed in ascending order:

5, 6, 7, 7, 9, 9, 10, 10, 13, 13, 13, 13, 18

a. What is the mean?

Median:

b. What is the median of the following data above?

Mode:

Percentile:

Quartile:

Interquartile range (IQR):

Ex: A data set has the following:

$25^{th}$ percentile = 7

$50^{th}$ percentile = 9.5

$75^{th}$ percentile = 13

## Mean (Formally):

Example: The height of dinosaurs is shown below:

| Height h(m) | $0 < h \leq 10$ | $10 < h \leq 14$ | $14 < h \leq 18$ | $18 < h \leq 22$ |
|---|---|---|---|---|
| Frequency | 5 | 2 | 1 | 2 |

What is the mean dinosaur height?

## SAMPLING AND OUTLIERS

## Discrete data:

## Continuous data:

## Sample:

## Reliable data:

## Sampling techniques:

A) Simple Randon — Equal chance of choosing out of a hat, number generator, etc.
Ex. Poll students from school-# assigned to Students, choose with random # generator.

B) Convenience — Choose easiest people to sample: ask your friends.
Problems? May not be representative of population.

C) Systematic — Choose random starting point, use fixed interval.
Ex. Make a list of all students in a class, choose every 3rd student...

| | |
|---|---|
| D) Quota | Sample sizes in proportion to who you're polling<br>Ex. 55% girls, 45% boys in school, so sample should have those same %. |
| E) Stratified | Spilt into strata (smaller groups)<br>Ex. Choose half DP1, half DP2 students |